ModelIt!

Teacher's Guide

The Carbon Cycle: Where Does Your Breath Go?

How Carbon Travels Between Air, Plants, Soil, and Back Again

Quick Reference

Grade Level	Grade 05L2
NGSS Standard	5-LS1-1, 5-LS2-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 5-LS1-1, 5-LS2-1

Develop a model to describe the movement of matter among plants, animals, decomposers, and the environment. Support an argument that plants get the materials they need for growth chiefly from air and water.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a model showing how carbon cycles between atmosphere, plants, soil, and back.
Disciplinary Core Idea	**LS2.B Cycles of Matter and Energy Transfer** — Matter cycles between living and nonliving parts of an ecosystem through processes like photosynthesis and decomposition.
Crosscutting Concept	**Systems and System Models** — Students model the carbon cycle as a system with feedback loops where outputs become inputs.

CAST Testing Alignment

MULTIPLE CHOICE:

If all sunlight were blocked for a month, what would happen to CO2 levels in the atmosphere?

CONSTRUCTED RESPONSE:

Explain how a carbon atom in the air you breathe out could end up in a tree, then in the soil, and then back in the air again.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

The carbon cycle is one of Earth's most important biogeochemical cycles. Carbon atoms move continuously between the atmosphere, living organisms, soil, and oceans in a complex loop that supports all life.

Carbon in the Atmosphere

Carbon dioxide makes up about 0.04% of Earth's atmosphere, but this small amount has an enormous effect on climate. Plants absorb CO2 during photosynthesis, and organisms release it during respiration and decomposition.

Photosynthesis as Carbon Capture

During photosynthesis, plants use sunlight energy to combine CO2 from the air with water to create glucose (sugar). This effectively pulls carbon out of the atmosphere and stores it in plant tissue.

Decomposition Completes the Cycle

When plants and animals die, decomposers (bacteria, fungi, worms) break down their bodies. This process releases the stored carbon back into the atmosphere as CO2, completing the cycle.

Soil as Carbon Storage

Soil holds more carbon than the atmosphere and all plants combined. Organic matter in soil — decomposed leaves, roots, and organisms — acts as a long-term carbon bank that can store carbon for centuries.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify CO_2 in Atmosphere, Sunlight Intensity, Photosynthesis Rate, Decomposition Rate, and Soil Carbon Storage as the five system components.

E **Establish Relationships**
Students determine that CO_2 and Sunlight positively affect Photosynthesis, that Photosynthesis negatively affects CO_2 (removes it), and that Decomposition positively affects CO_2 (releases it).

V **Visualize & Model**
Students build the carbon cycle model showing feedback loops where photosynthesis removes CO_2 and decomposition adds it back.

E **Evaluate Outcomes**
Students run the Sunlight Shutdown scenario and observe CO_2 building up when photosynthesis stops, then test the High CO_2 scenario to see plants respond by photosynthesizing faster.

R **Revise & Extend**
Students add Ocean Absorption or Fossil Fuel Burning to explore how humans and oceans affect the carbon cycle.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Every breath you take is part of a giant cycle. Where does it go?"*

Watch For

Have students breathe out onto their hands and ask: Where does that CO2 go?

Slide 2: Learning Objectives

2 min

Say: *"Today we trace carbon through the entire cycle."*

Watch For

Have students read objectives aloud and circle unfamiliar words.

Slide 3: Big Question

2 min

Say: *"This is the mystery we will solve with our model."*

Watch For

Quick-write: students write their best guess in 30 seconds.

Slide 4: LEVER Framework

2 min

Say: *"We will use all five LEVER steps to build and test our carbon cycle model."*

Watch For

Briefly review each step with the class.

Slide 5: Activity 1: Components

8 min

Say: *"Our model has five parts. Two are external — we control them. Three are internal — they respond."*

Watch For

Students sort component cards into external and internal categories.

Slide 6: Activity 2: Connections

8 min

Say: *"Notice that photosynthesis has a NEGATIVE effect on CO2 — it removes carbon from the air. This creates a feedback loop."*

Watch For

Guide students to draw all five relationship arrows in ModelIt.

Slide 7: Activity 3: Simulation

10 min

Say: *"What happens when we take away sunlight? Let's test it."*

Watch For

Students run Sunlight Shutdown scenario and record observations.

Slide 8: Discoveries

5 min

Say: *"Carbon moves in a loop, not a straight line. That's what makes it a cycle."*

Watch For

Class discussion: How is this different from an energy chain?

Slide 9: STEM Challenge

10 min

Say: *"Now use what you learned to design a carbon-capturing schoolyard."*

Watch For

Distribute design materials, circulate to assist groups.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** CO_2 in Atmosphere, Sunlight Intensity
- **INTERNAL:** Photosynthesis Rate, Decomposition Rate, Soil Carbon Storage

Relationships

- **CO_2 in Atmosphere to Photosynthesis Rate:** POSITIVE (+) — More CO_2 gives plants more raw material for photosynthesis.
- **Sunlight Intensity to Photosynthesis Rate:** POSITIVE (+) — More sunlight provides more energy for plants to photosynthesize.
- **Photosynthesis Rate to CO_2 in Atmosphere:** NEGATIVE (-) — Faster photosynthesis removes more CO_2 from the air.
- **Decomposition Rate to CO_2 in Atmosphere:** POSITIVE (+) — Faster decomposition releases more stored carbon back as CO_2.
- **Photosynthesis Rate to Soil Carbon Storage:** POSITIVE (+) — More photosynthesis creates more plant matter that eventually stores carbon in soil.

Key Discoveries

- Carbon moves in a CYCLE, not a one-way path — it loops between air, plants, soil, and back
- Photosynthesis removes CO_2 from the atmosphere, acting like a natural filter
- Decomposition releases stored carbon back into the atmosphere as CO_2
- When sunlight disappears, photosynthesis stops and CO_2 builds up in the air

Multiple Choice Post-Assessment Answer Key

1. In the carbon cycle model, what happens when Sunlight Intensity is locked at 0%?

Correct: C) CO_2 in Atmosphere rises because photosynthesis stops but decomposition continues releasing CO_2

Correct! Without sunlight, photosynthesis cannot occur, so no CO_2 is being removed from the atmosphere. Meanwhile, decomposition continues releasing stored carbon as CO_2, causing atmospheric levels to rise.

2. What type of relationship exists between Photosynthesis Rate and CO_2 in Atmosphere?

Correct: B) Negative: more photosynthesis removes CO_2 from the air

That's right! Photosynthesis absorbs CO_2 from the atmosphere and converts it into sugar. Faster photosynthesis means more CO_2 is removed. This is a negative relationship.

3. In the model, when CO_2 in Atmosphere is set to 90%, what happens to Photosynthesis Rate and why?

Correct: B) Photosynthesis Rate increases because plants have more CO_2 raw material to use

Correct! CO_2 is a raw material for photosynthesis. When more CO_2 is available, plants can photosynthesize faster. This is a positive relationship between CO_2 and Photosynthesis Rate, and it creates a negative feedback loop that pulls CO_2 back down.

4. Why is the carbon cycle called a 'cycle' and not a 'chain'?

Correct: C) Because carbon atoms loop continuously between the atmosphere, plants, soil, and back again

Exactly! Carbon is never created or destroyed. It moves from the atmosphere into plants through photosynthesis, into the soil when organisms die, and back to the atmosphere through decomposition. The same carbon atoms cycle over and over.

5. A scientist notices that a forest has very little dead organic matter on the ground despite many trees dropping leaves. What does this tell you about the carbon cycle in this forest?

Correct: B) Decomposition Rate is very high, breaking down dead matter quickly and releasing carbon back to the atmosphere

That's right! If trees are dropping leaves but the ground is clear, decomposers must be working very quickly. Warm, moist conditions speed up decomposition, breaking down dead matter and releasing stored carbon as CO2 back into the atmosphere.

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Plants get carbon from soil"	Plants get carbon from CO_2 in the AIR through photosynthesis, not from the soil. Soil provides minerals and water.
"Carbon is only in living things"	Carbon exists in the atmosphere (CO_2), in oceans (dissolved CO_2), in rocks (limestone), and in fossil fuels. Living things are just one stop on the cycle.
"Decomposition is bad"	Decomposition is essential for the carbon cycle. Without it, dead matter would pile up and nutrients would never return to the soil for new plants.